

Building Data Teams

Teamwork, Kanban & Your Group Project



Roadmap

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Teamwork

Teamwork can be challenging

Most people think working in groups is easy. It isn't - not even for us, coordinating this course!
Let's do an exercise that proves the point.

1 | Teamwork

The Marshmallow Challenge

Objective

- Build the tallest freestanding structure, measured from the table to the top of the marshmallow.

Materials & rules

- Use only what you're given: spaghetti, string, tape
- The marshmallow must sit on top
- Destroying or eating it disqualifies your team

Time & prize

- 15 minutes, timed against a countdown video. Winning team gets a prize!

Some questions for you

- Was there a leader on your team?
- Who decided who the leader would be?
- If you had no leader, would designating one have helped?
- Did you have a plan?

Some lessons: who performs how

Perform poorly

- Business-school teams often do worst
- They spend most of their time agreeing on ONE plan
- They run out of time rushing that plan - and it collapses



Perform well

- Kindergarteners are surprisingly good at this
- They skip planning and just start building
- Many attempts collapse, but each one teaches them something



More lessons

Prototype early

- The exercise rewards iterative design over one big upfront plan.

It's heavier than it looks

- The marshmallow is heavier than most people assume - budget for it structurally, not just visually.

Beware hidden assumptions

- Teams that add the marshmallow last, assuming it's light, often watch their structure collapse. Every project has its own hidden 'marshmallow.'

Everything exciting at a company comes from projects - and every project succeeds or fails as a group, not as individuals. You will never work alone.

2

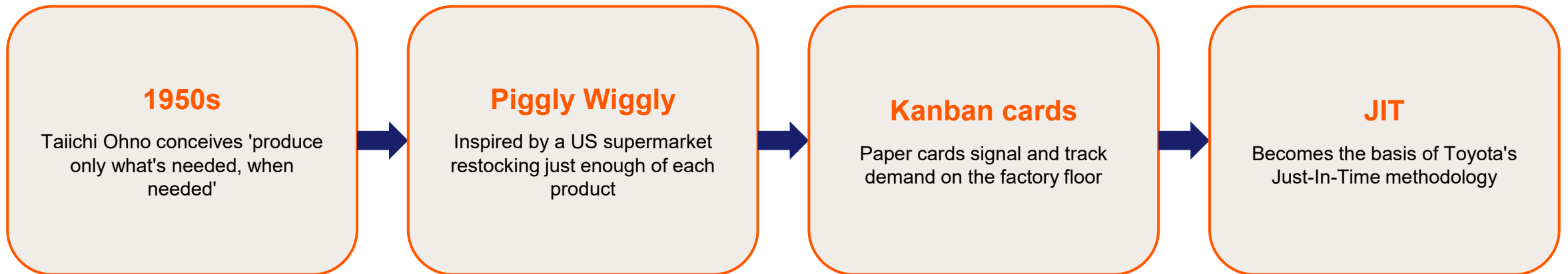
Kanban

What is Kanban?

Kanban is a visual way to manage work as it flows through a process: every task is visible on a board, you cap how much is in progress at once, and you keep improving the flow. Originally a factory-floor method, today it runs everything from software teams to your group project.

► Want the full guide? [Kanban, explained \(Atlassian\)](#).

Where Kanban came from



2 | Kanban

Core values of Kanban

Flow

- Continuously and predictably deliver value
- Deliver often, in small increments

Visibility

- Make work and its workflow visible, so issues surface early
- Prioritize openly, so the team sees what matters most

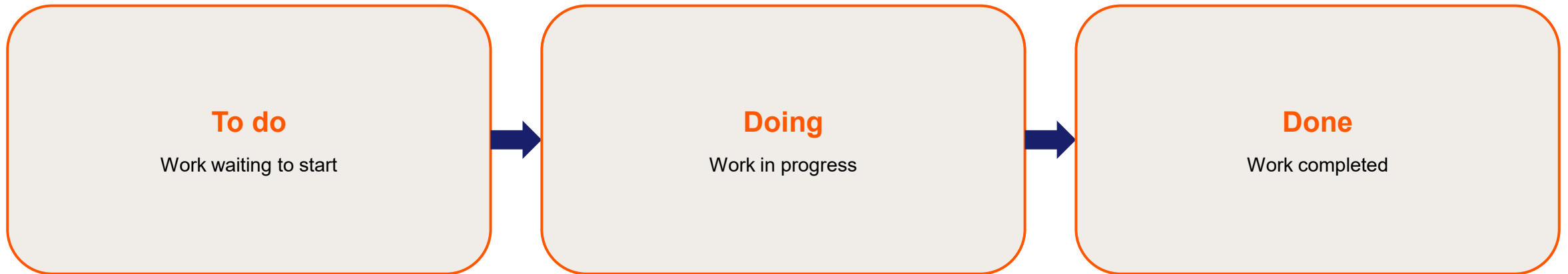
Limit WIP

- Cap how much work is in progress at once
- Less multitasking - shorter cycle times, higher quality

Limit Work in Progress

- Reduces cycle time per task
- Ensures WIP is the highest-priority task
- Reduces or eliminates queues
- Reduces multi-tasking by team members
- Rule of thumb: keep WIP at 5 or fewer items per person

Kanban boards



2 | Kanban

It's all about the cards

One card, one task

- Every work item is a separate card, tracking progress visually as it moves through the workflow.

What a card shows

- Who's responsible
- A short description of the task
- How long it's estimated to take

The goal: reach Done

- Each card moves from To Do through every column until it lands in Done - once every card is there, the project is finished.

You won't nail your board's setup on the first try - you'll add or remove columns, adjust cards, and rearrange priorities as you go. (SpaceX didn't land its first rocket booster either.)

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Group Project

Your deliverables

- Written Report (citations tracked via Zotero)
- Video Presentation, submitted in Canvas
- Kanban Board in Trello
- The Zotero library you used for the project
- Full detail: Canvas -> Group Project module

3 | Group Project

The Written Report

Methodology

- Explain neural networks to a client who knows nothing about them - no complicated math.

Application

- Real-world examples of neural networks applied to your assigned industry.

References

- Every source you used, tracked in Zotero.

The Video Presentation

Submit a video presentation of your report - maximum 10 minutes. Use animation tools, record yourselves, whatever works. Be creative - try to impress the client.

3 | Group Project

Building your Kanban board

Define your flow

- At least 3 columns - To Do, Doing, Done.
Add more if your work needs them (e.g. an Editing column).

Set a WIP limit

- Rule of thumb: 1.5x the number of people who can work a step at once. Doesn't apply to To Do or Done.

Split big tasks

- Anything over a couple of days, or needing multiple people, becomes several smaller cards you can track independently.

Grading

- Description of Neural Networks methodology - 2 pts
- Application to the assigned industry - 2 pts
- Use of the Kanban Board - 2 pts
- Presentation (instructor-assessed) - 3 pts
- Correct use of references (Zotero) - 1 pt
- Total - 10 pts

Due dates & group meetings

Due dates

- See Canvas for the exact submission deadline



Group meetings

- Ask any time - don't leave it to the last minute
- I'll check in with a different group each week



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Trello & Zotero

Trello setup

You'll share a Trello board with me as your virtual Kanban board - your project's To Do / Doing / Done, online. A short setup screencast is on the way; until then, the official Trello guide below walks you through creating your first board.

► New to Trello? [Trello Guide: set up your first board \(official\)](#).

Zotero setup

You'll track your Group Project's citations and references here, synced with MS Word so your bibliography builds itself. A short setup screencast is on the way; until then, the official Zotero guide below covers the essentials.

► New to Zotero? [Zotero Quick Start Guide \(official\)](#).

This week: how to run a Kanban board, manage it with Trello, and track your references with Zotero - the exact toolkit you'll use for your Group Project all semester.



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